

2024



Coinbase Launches an Ultra-Low-Latency Cryptocurrency Exchange on AWS

Learn how Coinbase launched the Coinbase International Exchange on AWS.

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Scales

to handle 100,000 messages per second

Offers stable and reliable

latency for fast trading decisions

Supports effective disaste

and consistent service



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Cryptocurrency exchanges need to operate with ultra-low latency so that traders can conduct transactions as fast as possible, a key factor in a market where milliseconds can mean the difference between profit and loss. Using Amazon Web Services (AWS), [Coinbase](#) addressed the challenge of maintaining ultra-low latency when building the Coinbase International Exchange.

Launched in May 2023, the Coinbase International Exchange is a cryptocurrency exchange that operates 24 hours per day, 7 days per week. It is regulated by the Bermuda Monetary Authority and has traded more than 15 billion dollars in volume. The exchange's core systems are built to handle 100,000 messages per second.



Achieving Sub-Millisecond Latency on Coinbase's Trading System

The Coinbase International Exchange was built on AWS from the ground up, using Cloud Native principles, powered by AWS managed services such as [Amazon Aurora](#), a relational database service that provides scalable performance and availability at a global scale. The Coinbase International Exchange uses Amazon Aurora for its high-speed, low-latency database needs, facilitating quick and reliable near real time and historical transaction lookup. These capabilities are critical for serving the needs of the trading ecosystem.

Additionally, Coinbase uses [Amazon Elastic Compute Cloud](#) (Amazon EC2) [z1d Instances](#)—which provide high compute performance, high memory, and fast disks—to manage its demanding workloads. The company chose these instances for their high compute performance, substantial memory, and, most importantly, their high-speed NVMe instance storage. By adopting NVMe storage on Amazon EC2 z1d instances, Coinbase benefits from fast disk write speeds. This helps the Coinbase Trading Engine process robust RAFT based trading logic and provide ultra-low trading latency services, which is vital to the cryptocurrency trading market. Coinbase uses these instances inside [Amazon EC2 cluster placement groups](#) to achieve the low-latency network performance necessary for tightly coupled node-to-node communication in the cluster.

Benefits of Using AWS

AWS services provide the performance, scalability, reliability, and developer-friendly environment that the Coinbase International Exchange needs to operate effectively at scale. Using high-performance cloud computing services, Coinbase can scale to manage high trading volumes, effectively meeting market demands. The



operational resilience and consistency are bolstered by robust infrastructure on AWS, offering stable and reliable latency crucial for fast trading decisions.

Furthermore, the AWS global network and replication features support effective disaster recovery, helping Coinbase maintain consistent service across various regions. On AWS, its developers can create, test, and deploy new features and updates as needed, driving innovation and improving responsiveness to market changes.”

The Coinbase International Exchange was born on the AWS Cloud, developed under Cloud Native Principles, and is going to continue to thrive on the AWS Cloud,” says Yucong Sun, senior staff engineer at Coinbase. “We couldn’t do that without powerful collaborations between the Coinbase team and the AWS support team, who helped us learn about the different features that we can use, as well as the best practices that we can do on the cloud.”

Coinbase at AWS re:Invent 2023

On November 27, 2023 at AWS re:Invent, Coinbase took center stage during an insightful breakout session. If you missed this informative breakout session or want to revisit the valuable lessons shared, you can [watch the full recording here](#).

About Coinbase

Crypto creates economic freedom by ensuring that people can participate fairly in the economy, and Coinbase is on a mission to increase economic freedom for more than 1 billion people. They are updating the century-old financial system by providing a trusted platform that makes it easy for people and institutions to engage with crypto assets, including trading, staking, safekeeping, spending, and fast, free global transfers. Coinbase also provides critical infrastructure for onchain activity and support builders who share our vision that onchain is the new online. And together with the crypto community, they advocate for responsible rules to make the benefits of crypto available around the world. The Coinbase International Exchange is only available to non-US institutions in select jurisdictions. US institutions are strictly prohibited from access. Crypto-derivatives are not available to retail clients based in the United Kingdom.

AWS Services Used

Amazon Aurora

Unparalleled high performance and availability at global scale with full MySQL and PostgreSQL compatibility

[Learn more »](#)



Amazon EC2 z1d Instances

Delivering a sustained all core frequency of up to 4.0 GHz, and a high memory to vCPU ratio

[Learn more »](#)

Amazon EC2 Cluster Placement Groups

To meet the needs of your workload, you can launch a group of interdependent EC2 instances into a placement group to influence their placement.

[Learn more »](#)

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